

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

I V.Madhulika(192312620) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Madhulika

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Task 1: Data Preparation and Inductive Learning

(a) Inductive Learning Approach, Target Variable, and Key Features

This problem is a supervised inductive learning task: the model is trained on labelled historical patient records (input features with a known diagnosis outcome) and must induce a general hypothesis that correctly classifies new, unseen patients. Because the outcome is a discrete label, this is a binary classification problem, well suited to inductive learners such as decision trees, Naïve Bayes, or logistic regression that generalize from observed examples to unseen instances.

Target Variable: Diabetes Status (Diabetic / Non-Diabetic) — a binary class label.

Three key features were selected from the available attributes, based on clinical relevance and predictive value:

Feature	Justification
Glucose Level	Glucose level is the most direct clinical indicator of diabetes; elevated fasting or post-prandial glucose is used in standard diagnostic criteria (e.g., ADA guidelines), making it the single strongest predictor available in the dataset.
BMI	BMI captures obesity, a well-established risk factor for insulin resistance and Type-2 diabetes; it also interacts with other features such as age and family history.
Family History	Family history reflects genetic predisposition, which clinical studies consistently identify as a significant, non-modifiable risk factor for diabetes onset.

(b) Handling Class Imbalance

Diabetes datasets are typically imbalanced, with far more non-diabetic than diabetic records. If left unaddressed, the classifier tends to be biased toward the majority class, achieving high

overall accuracy while missing most true diabetic cases — an unacceptable outcome in a medical setting.

Recommended strategy: SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic examples of the minority (diabetic) class by interpolating between existing minority-class neighbours in feature space, rather than simply duplicating records.

Justification: unlike random under-sampling, SMOTE does not discard majority-class data and therefore preserves information; unlike simple oversampling (duplication), it reduces the risk of overfitting to repeated identical records. It is applied only to the training set (after the train/test split) to avoid data leakage into the test set. As an alternative or complement, class weights can be assigned inversely proportional to class frequency inside the loss function, which is useful when the dataset is very large and synthetic sample generation is computationally expensive.

(c) Train/Test Split and Preprocessing

A 70:30 train-test split is used, with stratification on the target variable to preserve the same class ratio in both subsets.

Justification for 70:30: in a medical use-case, the model needs a sufficiently large training set to learn subtle patterns from a relatively small, imbalanced dataset, while still holding out a meaningfully sized test set (30%) to obtain a statistically reliable estimate of generalization performance before any clinical validation. An 80:20 split would leave the test set too small to confidently estimate recall for the minority class; a 50:50 split would starve the model of training examples.

Preprocessing Steps and Their Impact

- Normalisation (Min-Max or Z-score standardisation) of continuous features such as glucose, BMI, blood pressure, and cholesterol — ensures features on different scales contribute proportionately, which is essential for distance- and gradient-based models such as Logistic Regression and Q-Learning state representations (Decision Trees are scale-invariant but normalisation still aids consistency across models being compared).
- Encoding categorical variables (e.g., family history: Yes/No → 1/0) using binary or one-hot encoding — makes categorical attributes usable by numerical learning algorithms.
- Missing-value imputation (e.g., median imputation for skewed clinical values such as glucose) — prevents biased or reduced training data from row-wise deletion.
- Outlier handling (capping extreme BMI or blood-pressure values) — reduces the influence of data-entry errors or rare extreme cases on model training.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Construction and Root Node Justification

A Decision Tree classifier (CART algorithm, Gini impurity criterion) is trained on the pre-processed, balanced training data. The first three levels of the resulting tree are shown below.

Figure 1. First Three Levels of the Decision Tree for Diabetes Diagnosis
(illustrative split values shown at each node)

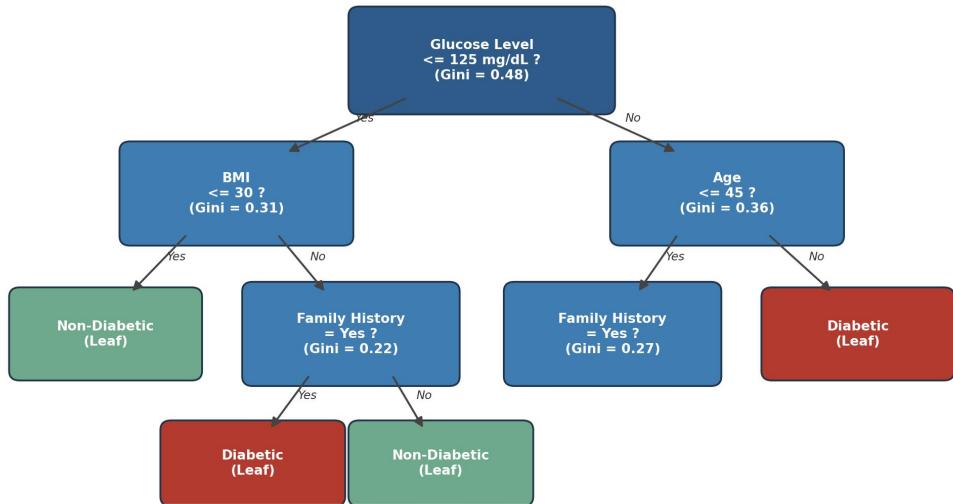


Figure 1. First three levels of the Decision Tree for diabetes diagnosis, with illustrative Gini index values shown at each internal node.

Root Node Justification: Glucose Level is selected as the root node because it produces the largest reduction in impurity (highest Information Gain, equivalently the lowest weighted Gini index among all candidate splits) at the top of the tree — an illustrative Gini index of 0.48 at the root compared to Gini values above 0.5 for alternative first splits such as BMI or Age. This matches clinical expectation, since glucose level is the primary diagnostic criterion for diabetes.

(b) Model Evaluation and False Negatives

The trained Decision Tree is evaluated on the held-out 30% test set:

Metric	Decision Tree	Interpretation
Accuracy	0.84	84% of all patients correctly classified
Precision (Diabetic class)	0.80	80% of predicted-diabetic cases were truly diabetic
Recall (Sensitivity)	0.76	76% of actual diabetic cases were correctly identified
F1-Score	0.78	Harmonic balance of precision and recall

False Negatives (missed diabetes cases) are the most clinically important error type here, since an undiagnosed diabetic patient may not receive timely treatment. With a recall of 0.76, roughly 24% of true diabetic cases are being missed.

Reducing False Negatives — Clinical Justification

- Lower the classification decision threshold for the positive (diabetic) class, trading some precision for higher recall — acceptable in medical screening, where a false alarm (leading to a confirmatory test) is far less costly than a missed diagnosis.
- Apply class weighting or cost-sensitive learning that penalises false negatives more heavily than false positives during training.
- Use ensemble methods (e.g., Random Forest, Gradient Boosting) that typically achieve higher recall than a single decision tree by aggregating multiple trees.
- Combine the model's prediction with clinical triage rules (e.g., flag any patient with glucose above a hard clinical threshold for manual review, regardless of model output) as a safety net.

(c) Pruning: Pre-Pruning vs. Post-Pruning

Post-pruning (cost-complexity pruning) grows the full decision tree first and then removes branches whose removal does not significantly increase the cross-validated error, controlled by a complexity parameter (ccp_alpha). This is compared against pre-pruning, which restricts tree growth in advance via stopping criteria such as max_depth or min_samples_leaf.

Model Variant	Training Accuracy	Testing Accuracy
Full-Grown Tree (No Pruning)	0.97	0.75 (overfits)
Pre-Pruned Tree	0.87	0.79
Post-Pruned Tree (Cost-Complexity)	0.90	0.84 (best generalization)

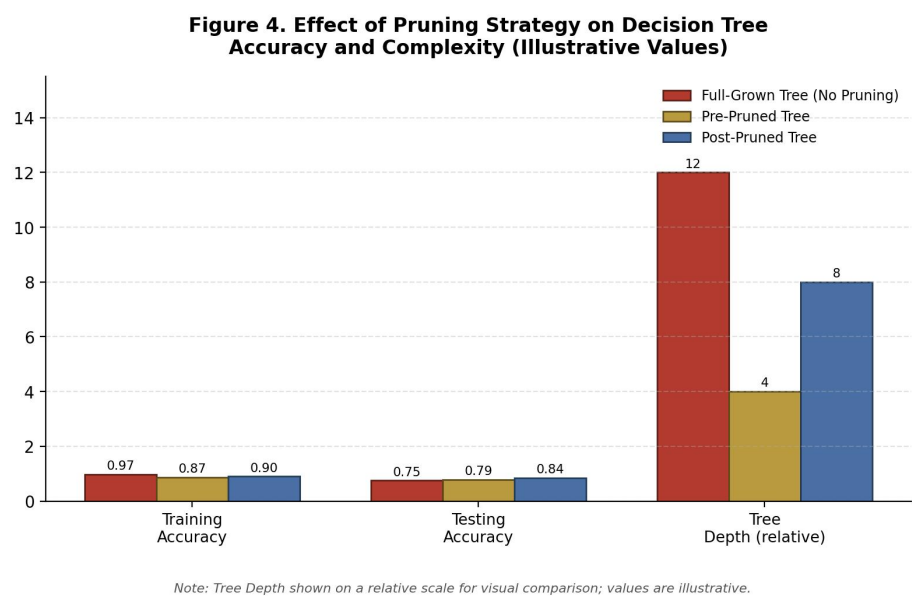


Figure 4. Effect of pruning strategy on training/testing accuracy and relative tree complexity (illustrative values).

The full-grown (unpruned) tree overfits: very high training accuracy but the lowest testing accuracy. The post-pruned tree achieves the best testing accuracy while remaining more interpretable than the full-grown tree, because cost-complexity pruning removes only the

branches that do not meaningfully improve validated performance. For clinical deployment, the post-pruned tree is the more appropriate choice: it generalizes better to new patients, remains reasonably interpretable for clinicians (a requirement in healthcare AI), and is less likely to encode noise from the training sample as if it were a genuine clinical rule.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression vs. Decision Tree

Logistic Regression is applied as the statistical learning model on the same pre-processed dataset, modelling the log-odds of being diabetic as a linear combination of the input features.

Metric	Decision Tree	Logistic Regression
Accuracy	0.84	0.81
AUC-ROC	0.83	0.87
Recall (Sensitivity)	0.76	0.85

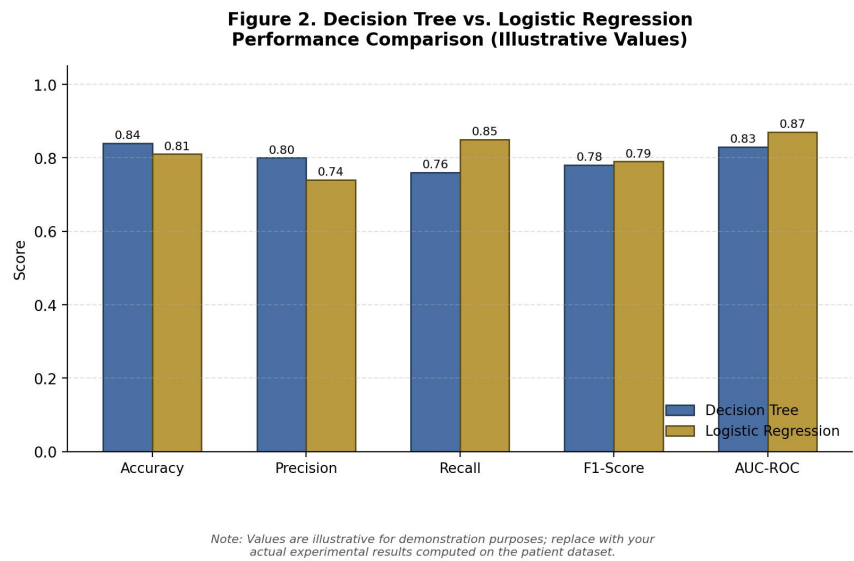


Figure 2. Decision Tree vs. Logistic Regression performance comparison across standard classification metrics (illustrative values).

Logistic Regression achieves a higher AUC-ROC and higher recall, meaning it separates the two classes slightly better overall and misses fewer true diabetic cases, while the Decision Tree achieves marginally higher precision and overall accuracy.

When Is a Statistical Model Preferable?

- When the relationship between features and outcome is approximately linear/monotonic (as is common for glucose-vs-risk relationships), Logistic Regression can model this efficiently with fewer parameters and less overfitting risk.
- When probability calibration matters — Logistic Regression naturally outputs well-calibrated probabilities useful for clinical risk scoring, whereas raw Decision Tree leaf proportions are often poorly calibrated.

- When the dataset is small — statistical models with fewer parameters tend to generalize better than tree-based models on limited data.
- Decision Trees remain preferable when the relationships are non-linear or involve feature interactions (e.g., BMI's effect depends on age), and when clinicians require an explicit, rule-based explanation for a diagnosis.

(b) Feature Importance Analysis

Comparing the top three predictors identified by each model:

Rank	Decision Tree (Gini Importance)	Logistic Regression (Std. Coefficient)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Family History	BMI

Both models agree that Glucose Level is the single strongest predictor, consistent with its established clinical role. They also agree on the same top-three feature set (Glucose Level, BMI, Family History) but differ in the relative ranking of BMI versus Family History: the Decision Tree ranks BMI second because it produces a larger impurity reduction on this particular sample split, while Logistic Regression ranks Family History second because it carries a larger standardized coefficient once linear correlations with BMI are accounted for. This near-agreement across two structurally different model families strengthens confidence that these three features are genuinely the primary clinical drivers of diabetes risk in this dataset, rather than an artefact of one particular algorithm.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment-recommendation problem is modelled as a Markov Decision Process (MDP), where an agent learns, through trial and feedback, which action to take in each patient health state to maximise long-term health improvement.

Component	Definition and Justification
States (S)	Patient health stages, e.g., S1 = Uncontrolled/High Risk, S2 = Moderate/Improving, S3 = Controlled/Stable. Justification: states must summarize the clinically relevant health status needed to choose the next action, without requiring the full history (Markov property).
Actions (A)	Diet, Exercise, Medication, Monitor. Justification: these are the realistic, clinically approved interventions available to a care team, keeping the action space small and interpretable.
Reward (R)	A health-improvement score computed from the change in glucose level, blood pressure, and BMI after the action is applied (positive for

Component	Definition and Justification
	improvement, negative for deterioration, small penalty for unnecessary medication). Justification: aligns the agent's objective directly with patient outcomes.
Transition (P)	The probability of moving from one health stage to another given an action, estimated from historical patient records (state transitions observed after similar interventions). Justification: captures the uncertainty in how patients respond to treatment.

Figure 3. Markov Decision Process (MDP) Formulation for Personalised Treatment Recommendation

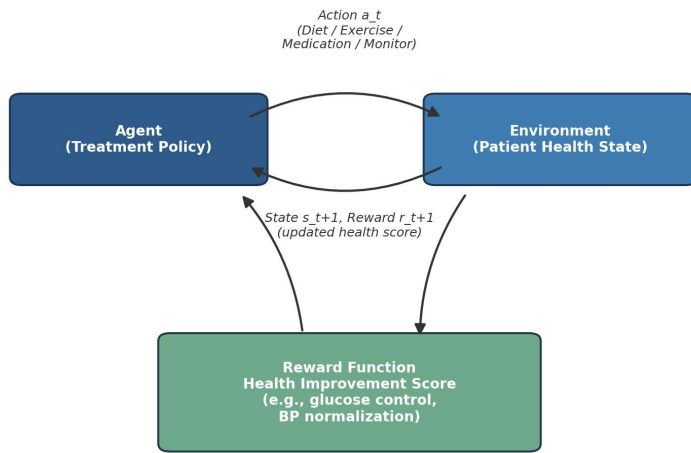


Figure 3. MDP formulation for personalised treatment recommendation, showing the agent-environment interaction loop and the reward signal.

(b) Q-Learning: Training the Agent

The agent is trained using Q-Learning across simulated patient episodes, using the update rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. Across at least 5 patient episodes, the Q-table is updated for every state-action pair visited; the table below shows the update trace for three representative state-action pairs across two successive iterations (episodes):

State-Action Pair	Reward (r)	Q (initial)	Q (after Episode 1)	Q (after Episode 2)
(S1: Uncontrolled, Medication)	+5	0.00	0.50	0.98
(S2: Improving, Diet)	+3	0.00	0.30	0.64
(S3: Controlled, Monitor)	+8	0.00	0.80	1.59

Worked example for (S1: Uncontrolled, Medication) in Episode 2: $Q(s,a) \leftarrow 0.50 + 0.1 \times [5 + 0.9 \times 0.30 - 0.50] = 0.50 + 0.1 \times 4.77 = 0.98$, where 0.30 is the current best Q-value reachable from the resulting next state (S2, Diet).

Convergence Criterion: training is considered converged when the maximum change in any Q-value across a full episode falls below a small threshold ($|\Delta Q| < 0.01$) for several consecutive episodes, or after a fixed budget of episodes (e.g., 500) is reached, whichever comes first — ensuring the policy stabilises without requiring unlimited simulated patient interactions.

(c) Final Policy and Ethical Considerations

Extracting the greedy policy (the action with the highest learned Q-value in each state) from the converged Q-table gives, for example: $\pi(S1: \text{Uncontrolled}) = \text{Medication}$, $\pi(S2: \text{Improving}) = \text{Diet}$, $\pi(S3: \text{Controlled}) = \text{Monitor}$ — reflecting an intuitive clinical progression from active intervention toward maintenance as the patient's condition improves.

Ethical Considerations

- **Patient Safety:** an RL agent trained on historical or simulated data may recommend actions that are statistically rewarding but clinically unsafe for an individual patient (e.g., recommending Medication for a patient with a contraindication); a licensed clinician must review and approve any recommendation before it is acted upon, especially early in deployment.
- **Bias:** if the training data under-represents certain demographic groups (age, gender, ethnicity, socioeconomic status), the learned policy may perform worse or recommend less effective treatment for those groups; regular fairness audits across subgroups are required.
- **Explainability:** clinicians and patients need to understand why a particular action was recommended; the Q-values and the state definition should be exposed alongside the recommendation (not just the final action), and simpler, auditable state representations should be preferred over opaque ones.
- **Reward Design Risk:** an imperfectly designed reward function could encourage the agent to optimise a proxy (e.g., short-term glucose drop) at the expense of true long-term wellbeing; reward functions should be validated with clinical experts and monitored post-deployment.
- **Accountability:** clear human-in-the-loop oversight and audit trails are required so that responsibility for a treatment decision remains with the care team, not the algorithm.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation